

Data Issues Report

Every data quality problem found across the 3 source files, and exactly what was done about it. Full detection/fix logic lives in `scripts/01_merge_pipeline.py`; a machine-readable version of this list is generated at `docs/data_issues_log.json` each time the pipeline runs.

Cross-source problem: no single ID field

None of the 3 files share a common identifier. - `source1_naukri_applicants.csv` has **email + phone** - `source2_gig_workers.csv` has **email only** (no phone) - `source3_cbnexus_contacts.csv` has **phone only** (no email)

Fix: used `source1` as a bridge table. Built a union-find (disjoint-set) structure that merges any two rows sharing a normalized phone OR a normalized email into one `person_id`. `source2` rows link to `source1` via email; `source3` rows link to `source1` via phone. Deliberately did **not** match on name — see the name-collision issue below for why that would have been wrong.

source1_naukri_applicants.csv

Issue	Detail	Fix
Inconsistent units	Current <code>CTC</code> column mixes raw annual rupees (e.g. <code>417964</code>) with values already in LPA (e.g. <code>4.2</code>)	Any value > 100 treated as raw rupees and divided by 100,000; everything normalized into one <code>ctc_lpa</code> float column
Inconsistent phone formats	Numbers appear as <code>+9190000000254</code> , <code>90000000237</code> , <code>090000000287</code>	Stripped all non-digits, removed country code (<code>91</code>) or leading <code>0</code> , normalized to a clean 10-digit string

Issue	Detail	Fix
Inconsistent date formats	Applied Date uses at least 4 different formats: 24-07-2026 , 2026-08-08 , 07/13/2026 , 7 Jul 2026	Tried multiple strptime formats in sequence until one parses; normalized to ISO YYYY-MM-DD
City name inconsistency	GURGAON / gurugram / Gurgaon (same city, different casing); Bangalore / bangalore / Bengaluru (renamed city); Delhi / new delhi / Delhi NCR (same city)	Built a synonym map to one canonical name per city; also trimmed stray trailing whitespace (e.g. "Noida ")
Duplicate row within source	"Rohit Verma" and "R. Verma" appear as two separate rows with identical email and phone — same person entered twice with an abbreviated name	Detected via matching (email, phone) pairs; kept the first occurrence, dropped the duplicate
Skill list formatting	Skills stored as a comma-separated string with mixed casing between sources	Lowercased and de-duplicated into a sorted list for consistent matching against source2's skill tags

source2_gig_workers.csv

Issue	Detail	Fix
Completely blank row	One row in the file is entirely empty	Detected (all fields null) and dropped
Column-shifted / corrupted row	One row's values are rotated across columns — its email_id field contains "react, javascript, mysql" (clearly a skills list, not an email), while worker_name contains what looks like an email address	Validated email_id against an email-format regex; when it fails, checked whether rotating the row's fields recovers a record identical to an existing clean row for the same worker. It did — this is a corrupted duplicate export of "Isha Chopra", not a new person — so it was dropped rather than "fixed" and kept

Issue	Detail	Fix
Inconsistent units	<code>rate</code> column mixes hourly rates ("1415/hr") and monthly rates ("15k/month") with no way to compare them directly	Converted monthly rates to an estimated hourly rate assuming ~176 working hours/month (22 days × 8 hrs); all normalized into one <code>rate_hourly_inr</code> float column
Inconsistent casing (status)	<code>status</code> values appear as <code>Active</code> / <code>active</code> / <code>ACTIVE</code> / <code>Inactive</code> / <code>paused</code>	Lowercased and normalized to a consistent set (<code>active</code> / <code>inactive</code> / <code>paused</code>)
Inconsistent casing (email)	Several <code>email_id</code> values are in ALL CAPS or mixed case, which would break exact-match joins against source1's lowercase emails	Lowercased every email before using it as a join key
City name inconsistency	Same synonym/whitespace issues as source1 (<code>Bangalore</code> vs <code>Bengaluru</code> , trailing spaces, etc.)	Same canonical city map applied

source3_cbnexus_contacts.csv

Issue	Detail	Fix
Duplicated header row mid-file	The column header line (<code>Name</code> , <code>Phone Number</code> , <code>City</code> , <code>Verified</code> , <code>Projects Completed</code>) is repeated partway through the file, as if two exports were concatenated without cleanup	Detected the repeated header line and stripped it before parsing, so it doesn't get read in as a fake data row
Inconsistent casing (Verified)	<code>Verified</code> column mixes <code>Y</code> / <code>N</code> / <code>yes</code> / <code>No</code> / <code>Yes</code>	Normalized to a proper boolean (<code>True</code> / <code>False</code>)
Inconsistent phone formats		Same phone normalization applied

Issue	Detail	Fix
	Same issue as source1: 9000000268 , +91-9000000131 , 919000000260 all appear	(strip to clean 10-digit number)
ALL CAPS names	Several names appear fully capitalized (RITU SHARMA , RAHUL MALHOTRA) inconsistent with title-case entries elsewhere	Normalized to title case for display
Name collision — two different people, same name (planted trap)	"Arjun Mehta" appears twice with two different phone numbers (9000000131 and 9000000272)	This is not a duplicate — it's two distinct people who happen to share a name. Matching was deliberately restricted to phone/email only (never name), so these two correctly stayed as separate person_id s instead of being incorrectly merged into one

Result

- 41 valid rows from source1 (after dropping 1 in-source duplicate), 30 from source2 (after dropping 1 blank + 1 corrupted row), 30 from source3 (after removing the injected duplicate header)
- Entity resolution produced **60 unique persons**, of which **25 matched across 2 or more source files**
- Verified specifically that the planted name-collision case ("Arjun Mehta", also "Deepak Nair") stayed split into separate people, and that the planted duplicate-with-typo case ("Rohit Verma"/"R. Verma") correctly collapsed into one